«Kazakh-British Technical University»

Faculty of Information Technology

LABORATORY WORK 1

ERD DIAGRAM - INTERNATIONAL AIRPORT

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1. Introduction

The goal of this lab is to design a simple database diagram (ERD) for an international airport. I will set up the main tables, connect them, and define the basic rules for how the airport database works.

2. Identified Entities

Entity	Purpose
Airport	Stores airport information: name, country, state, city, creation and update timestamps.
Airline	Stores airline company information including airline code, name, country, creation and update timestamps.
Flight	Stores scheduled and actual flight information, gates, airline, departure/arrival airports, and timestamps.
Passenger	Stores passenger personal and passport information.
Booking	Stores a passenger booking for a flight, its status, platform, ticket price, and timestamps.
Booking_Changes	Separate entity used to record changes made to a booking flight. The detailed attribute set is a design assumption because the task only states that changes must be stored separately.
BoardingPass	Stores boarding pass information issued for a booking.
Baggage	Stores baggage registered during a booking, including weight and timestamps.
BaggageCheck	Stores the baggage checking result associated with a booking and passenger.
SecurityCheck	Stores airport security check information for a passenger.

3. Attributes, Keys and Data Types

Airport

Attribute	Type	Constraint	Description
airport_id	BIGINT	PK	Unique identifier.
airport_name	VARCHAR(150)	Required	Airport name.
country	VARCHAR(100)	Required	Country.
state	VARCHAR(100)	Optional	State/region.
city	VARCHAR(100)	Required	City.
created_at	TIMESTAMP	Required	Creation time.
updated_at	TIMESTAMP	Required	Last update time.

Airline

Attribute	Type	Constraint	Description
airline_id	BIGINT	PK	Unique identifier.
airline_code	VARCHAR(10)	Required / UQ	Airline code;
name	VARCHAR(150)	Required	Airline name.
country	VARCHAR(100)	Required	Country.
created_at	TIMESTAMP	Required	Creation time.
updated_at	TIMESTAMP	Required	Last update time.

Flight

Attribute	Type	Constraint	Description
flight_id	BIGINT	PK	Unique identifier.
departing_gate	VARCHAR(10)	Optional	Departure gate.
arriving_gate	VARCHAR(10)	Optional	Arrival gate.
airline_id	BIGINT	FK	References Airline.
departure_airport_id	BIGINT	FK	References Airport.
arrival_airport_id	BIGINT	FK	References Airport.
scheduled_departure_time	TIMESTAMP	Required	Scheduled departure time.
scheduled_arrival_time	TIMESTAMP	Required	Scheduled arrival time.
actual_departure_time	TIMESTAMP	Optional	Actual departure time.
actual_arrival_time	TIMESTAMP	Optional	Actual arrival time.
created_at / updated_at	TIMESTAMP	Required	Audit timestamps.

Passenger

Attribute	Type	Constraint	Description
passenger_id	BIGINT	PK	Unique identifier.
first_name	VARCHAR(100)	Required	First name.
last_name	VARCHAR(100)	Required	Last name.
gender	VARCHAR(20)	Required	Gender.
date_of_birth	DATE	Required	Date of birth.
country_of_citizenship	VARCHAR(100)	Required	Citizenship country.
country_of_residence	VARCHAR(100)	Required	Residence country.
passport_number	VARCHAR(30)	Required / UQ	Passport number;
created_at / updated_at	TIMESTAMP	Required	Profile timestamps.

Booking

Attribute	Type	Constraint	Description
booking_id	BIGINT	PK	Unique identifier.
flight_id	BIGINT	FK	References Flight.
status	VARCHAR(30)	Required	Booking status.
booking_platform	VARCHAR(50)	Required	Platform used for booking.

Attribute	Type	Constraint	Description
created_at / updated_at	TIMESTAMP	Required	Booking timestamps.
ticket_price	NUMERIC(10,2)	Required	Ticket price.
passenger_id	BIGINT	FK	References Passenger.

Booking_Changes

Attribute	Type	Constraint	Description
change_id	BIGINT	PK	Unique identifier for change record.
booking_id	BIGINT	FK	References Booking.
old_flight_id	BIGINT	FK (recommended)	Flight before change.
new_flight_id	BIGINT	FK (recommended)	Flight after change.
change_reason	VARCHAR(255)	Optional	Reason for change.
changed_at	TIMESTAMP	Required	Change time.

BoardingPass

Attribute	Type	Constraint	Description
boarding_pass_id	BIGINT	PK	Unique identifier.
seat	VARCHAR(10)	Required	Assigned seat.
boarding_time	TIMESTAMP	Required	Boarding time.
created_at / updated_at	TIMESTAMP	Required	Pass timestamps.
booking_id	BIGINT	FK	References Booking.

Baggage

Attribute	Type	Constraint	Description
baggage_id	BIGINT	PK	Unique identifier.
weight_kg	NUMERIC(6,2)	Required	Weight in kilograms.
created_date / updated_date	TIMESTAMP	Required	Baggage timestamps.
booking_id	BIGINT	FK	References Booking.

BaggageCheck

Attribute	Type	Constraint	Description
baggage_checking_id	BIGINT	PK	Unique identifier.
check_results	VARCHAR(30)	Required	Check result.
created_at / updated_at	TIMESTAMP	Required	Check timestamps.
booking_id	BIGINT	FK	References Booking.
passenger_id	BIGINT	FK	References Passenger.

SecurityCheck

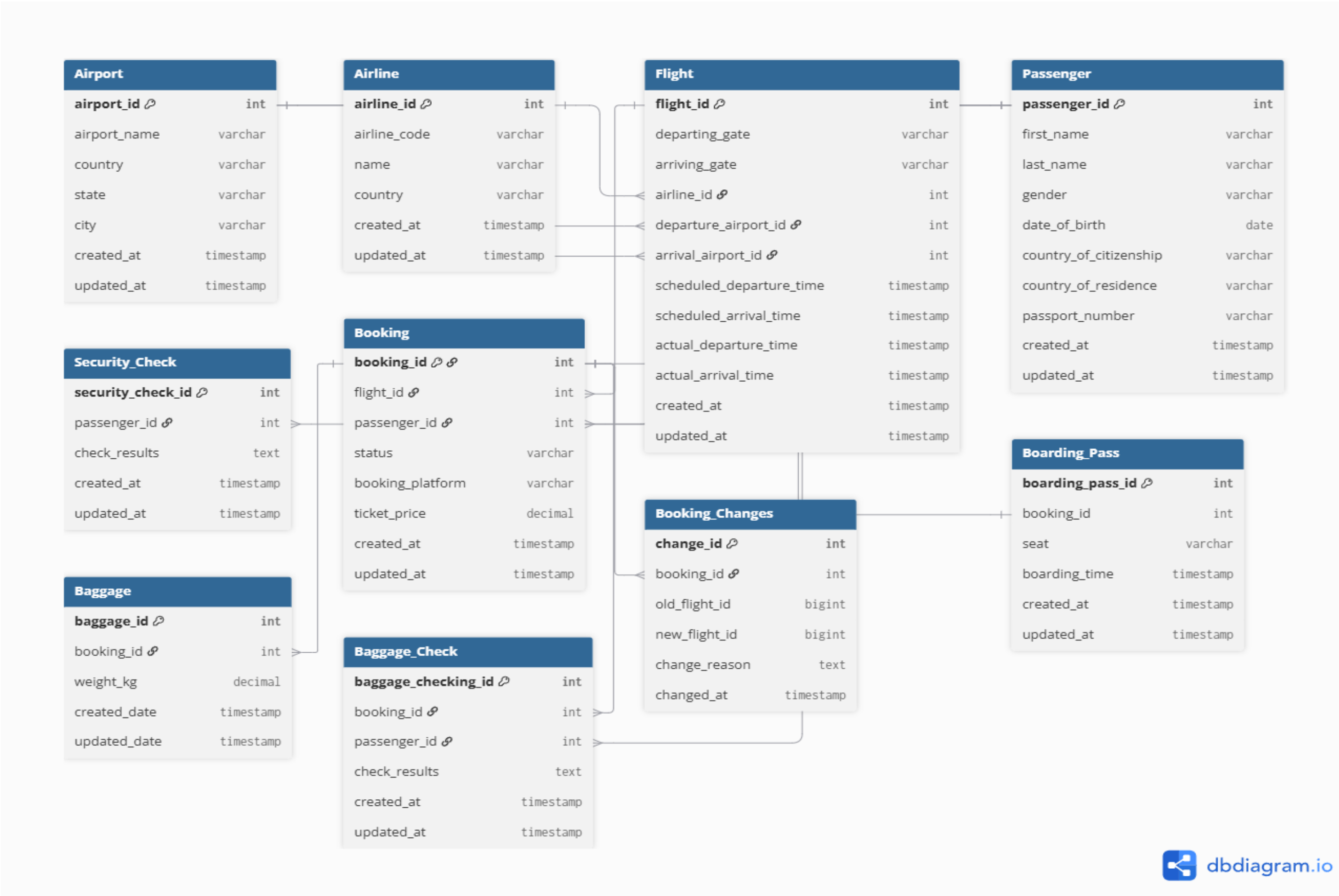
Attribute	Type	Constraint	Description
security_check_id	BIGINT	PK	Unique identifier.
check_results	VARCHAR(30)	Required	Security check result.
created_at / updated_at	TIMESTAMP	Required	Check timestamps.
passenger_id	BIGINT	FK	References Passenger.

4. Relationships and Cardinalities

Relationship	Cardinality	Explanation
Airport - Flight (departure)	1 : N	One airport can be the departure airport for many flights; each flight has one departure airport.
Airport - Flight (arrival)	1 : N	One airport can be the arrival airport for many flights; each flight has one arrival airport.
Airline - Flight	1 : N	One airline operates many flights; each flight belongs to one airline.
Passenger - Booking	1 : N	One passenger can make many bookings; each booking references one passenger.
Flight - Booking	1 : N	One flight can have many bookings; each booking references one flight.
Booking - BoardingPass	1 : 0..1	A booking may have zero or one boarding pass in the simplified model; each boarding pass belongs to one booking.
Booking - Baggage	1 : N	One booking can contain multiple baggage records; each baggage record belongs to one booking.
Booking - BaggageCheck	1 : N	A booking can have multiple baggage-check records; each record references one booking.
Passenger - BaggageCheck	1 : N	A passenger can have multiple baggage-check records; each record references one passenger.
Passenger - SecurityCheck	1 : N	A passenger can have multiple security checks; each security check belongs to one passenger.
Booking - Booking_Changes	1 : N	A booking can have multiple recorded flight changes; each change record belongs to one booking.

5. ER Diagram

The following ERD shows the proposed relational structure. PK means Primary Key, FK means Foreign Key, and UQ means Unique constraint.



6. ERD Legend / Notation

Notation	Meaning
PK	Primary Key - uniquely identifies a row.
FK	Foreign Key - references a primary key in another entity.
UQ	Unique constraint - prevents duplicate values.
1 : N	One-to-Many relationship.
1: 0..1	One-to-zero-or-one relationship.
Required	Attribute is expected to contain a value.
Optional / NULL	Value may be absent, especially for actual flight times before a flight operates.

7. Conclusion

Overall, this diagram shows how the airport database works. Primary and foreign keys make sure all the data stays accurate and connected. It splits everything into clean tables like passengers, baggage, and flights, making it easy to implement in PostgreSQL.